INSURANCE · AFTER-SALES SERVICE · SULTANATE OF OMAN

Rasd: an AI agent for after-sales service tickets

Rasd takes customer requests that come in after a policy is issued (ownership transfer, UAE extension, policy extension and more), checks the documents, talks to insurers by email or through their portals, and follows up on replies and reminders on its own. A staff member steps in at exactly two points: before a request goes to the insurer, and before a payment link goes to the customer.

The problem

Staff read, classify, chase documents, email insurers and follow up on every ticket by hand. Tickets run late and some are forgotten.

The saving

AI handles the repetitive work and the follow-ups, so each ticket takes less staff time and the same team handles more volume.

The safeguard

Nothing goes to an insurer and no payment link reaches a customer without approval from the responsible staff member. Every action is logged.

• AI work • Staff approval • Done • Alert or reminder

ARCHITECTURE

How the pieces fit together

Each ticket has a **LangGraph** graph that models its full journey. Temporal owns time: polling every 20 minutes, sending a reminder after a day, and resuming the graph when a reply arrives. Under both sits a set of tools the agent calls: OCR, RAG, email and Playwright.

Inbound channels

- WhatsApp (simulated)
- Customer email
- Website form

Every message becomes an event in the system

The brain: LangGraph

- Classify the request and extract data
- Check documents
- Draft the insurer request
- Read replies and prices
- Two approval gates (interrupt)

State persisted in a checkpointer

The scheduler: Temporal

- Poll inbox and portals
- Remind insurers automatically
- Remind customers about documents
- Escalate when overdue

Timers survive a server restart

Tools

- OCR for images and PDFs
- RAG over insurer procedures
- Email (IMAP / SMTP)
- Playwright for portals
- Payment link (test mode)

Data and screens

- PostgreSQL with pgvector
- Document storage
- Staff approval screen
- Stats dashboard
- Settings screen

STACK

What each technology does

`LangGraph` Brain

Decides **what** happens at each step: classifies the request, decides whether the documents are complete, drafts the request, pauses at both approval gates with `interrupt`, and resumes with `Command(resume=...)` once staff approve.

`Temporal` Recommended

Decides **when** things happen. Each ticket is a workflow with durable timers: poll every 20 minutes, remind after 24 hours, escalate after 3 days. A server restart loses no timer. **Why not RabbitMQ?** RabbitMQ is a message queue, great for taking in events, but it does not manage timers or long waits, so it would need a separate scheduler. It can be added later for inbound messages only.

OCR

Document checks

A PDF with a text layer is read directly. A photo or scanned PDF goes through OCR (Tesseract for Arabic and English, or a vision model). The LLM then turns the text into structured fields: name, civil ID, expiry date, plate number. Then **validation rules**: is it still valid? Does the name match? Does the plate match the policy?

RAG

Knowledge

A knowledge base of each insurer's procedures, the documents they require per service, their email templates, policy wordings and internal SOPs. The agent uses it to know **exactly what Insurer A needs for a transfer** and to answer customers from a trusted source.

`Playwright`

Portals

Some insurers have no API, only a web portal. Playwright signs in, fills the form, uploads the documents and returns periodically to check the request status. **It saves a screenshot of every step** as evidence. The prototype runs it against a mock portal.

`FastAPI · PostgreSQL`

Foundations

An API for tickets, approvals and settings, and a database for customers, policies, tickets, messages, prices and the audit log. `pgvector` in the same database serves RAG, so no extra store is needed.

TICKET STATES

From open to closed

New → Awaiting documents → In review → Sent to insurer → Awaiting insurer approval or quote → In review (payment) → Awaiting payment → Issuing → Closed

From any state a ticket can move to Escalated if it passes the deadline set for its service type, if the insurer declines, or if the AI's confidence in reading a document is low.

SCENARIOS

Three tickets, start to finish

Scenario 1 · Ownership transfer via portal

Ticket T-1042 · Insurer A: web portal + insurer approval required

Poll every 20 min · remind after 24 h

Sun 09:02 — Customer

"Hi, I sold my car and want to transfer the insurance to the buyer." Attached: a photo of the vehicle registration (mulkiya) and a PDF of the buyer's ID card.

09:02 — LangGraph · classify

Request type: **ownership transfer** (confidence 0.96). Finds the policy from the phone number and sees it was issued by **Insurer A**, whose channel is its portal.

09:03 — OCR + validation — RAG

Registration: ✓ **plate matches policy** Buyer ID: ✓ **valid until 2029**

Buyer's licence: ✗ **not attached**

From RAG: Insurer A requires the new owner's driving licence for transfers.

09:03 — Message to customer

"Thanks, we've received the registration and the buyer's ID. We just need the buyer's driving licence (photo or PDF) to complete the transfer."

State: awaiting documents. The request does not move until it is complete.

10:15 — Customer — OCR

Sends a photo of the licence. OCR reads it: valid until 2028, and the name matches the ID ✓. Documents are now complete.

10:16 — Draft request

The agent prepares every portal form field with the attachments. State: **in review**.

10:40 — ⏸ Staff approval 1

The staff member reviews the data and documents and clicks "Approve and send".

10:41 — Playwright

Signs in to Insurer A's portal, fills the form, uploads the documents and stores reference `A-77310` plus a screenshot of each step.

11:01 → … — Temporal · poll every 20 min

Checks the portal and the inbox. Status at the insurer: "Pending underwriting approval". Polling repeats quietly and nothing is pushed to staff.

Mon 10:41 — Automatic reminder

> 24 hours without a reply, so the system emails Insurer A's operations team on its own: "Following up on ownership transfer request A-77310 submitted yesterday. The customer is waiting for completion."

Reminders use a fixed template and need no approval. After a maximum of 3 reminders the ticket is escalated to staff.

13:20 — Read reply

The portal shows **Approved** with a transfer fee of `OMR 5.000`. The price matches the price table ✓.

13:35 — ⏸ Staff approval 2

The staff member checks the amount and the message text, then approves sending the payment link.

14:02 — Paid · Playwright

The customer pays. The system goes back to the portal, requests the endorsement, downloads it and sends it to the customer.

14:10 — Closed

The ticket closes and the customer is asked to rate the service. Actual staff time: about 4 minutes across both approvals.

Scenario 2 · UAE extension via two-round email

Ticket T-1057 · Insurer B: email, two-round pricing · Poll every 30 min · remind after 4 h

Tue 08:10 — Customer

"I'd like to add UAE cover to my insurance. I'm driving to Dubai on Thursday for two weeks."

08:11 — Classify · RAG

Request type: **UAE extension**, priority **high** because travel is in two days. The policy is comprehensive and active, and the registration is on file, so no documents are missing. From RAG: Insurer B quotes by email first, then needs a confirmation email after payment.

08:25 — ⏸ Staff approval 1

Staff review the quote-request email drafted by the agent and approve it.

08:26 — Email · round 1

Subject: `[T-1057] UAE extension quote request · Policy MV-2291`
The ticket number in the subject is what links the insurer's reply back to the right ticket.

08:56 → … — Temporal · poll every 30 min

Checks the inbox for a reply in the same thread.

11:10 — Read reply · Price changed

The insurer quotes `OMR 12.000`, but the price table says `OMR 10.000`. The agent does not change the table itself. It flags the ticket and suggests a price update.

11:22 — ⏸ Staff approval 2

Staff see the alert, accept the new price and update the table in one click (who changed it and when is recorded). Then they approve a payment link for OMR 12.000.

12:05 — Paid

The customer pays.

12:06 — Email · round 2

A confirmation email to Insurer B with proof of payment and a request to issue the endorsement. It completes a transaction staff already approved, so it goes out automatically. Each insurer's settings can make it require approval instead.

14:40 — Read reply · Closed

The endorsement PDF arrives. The agent checks that the dates and policy number are correct and sends it to the customer, two days before the trip.

Scenario 3 · Policy extension with a missing document

Ticket T-1063 · Insurer C: email + portal · Remind customer after 24 h · close after 3 reminders

Wed 17:30 — Customer

"I want to extend my insurance by two months." Attached: a PDF of the driving licence.

17:31 — OCR + validation

Licence: ✗ expired 2026-08-14 Registration: ✗ expired

The agent stops the process and works out exactly what is needed.

17:31 — Message to customer

"The licence you attached expired on 14 August, and our records show the vehicle registration has also expired. We need a valid licence and a renewed registration to complete the extension."

Thu 17:31 — Customer reminder 1

No reply within 24 hours, so the system sends a polite reminder.

Fri 09:12 — Customer · OCR · Low quality

Sends a photo of the renewed registration, but it is blurry. Read confidence is 0.58, below the threshold, so the agent asks for a clearer photo. The new licence comes through fine ✓.

09:40 OCR

A clear photo of the registration: valid until 2027 ✓. Documents are complete, so the ticket moves to "In review" and follows the normal path.

Otherwise If the customer never replies

After 3 reminders with no reply, the ticket closes as "No customer response" and staff are notified. It can be reopened at any time.

SETTINGS

Polling and reminders per service type

Every value is edited from the settings screen with no code change. The figures below are example defaults.

Service	Poll inbox and portal	Remind insurer after	Max reminders	Remind customer about documents	Escalate to staff after
Ownership transfer	every 20 min	24 h	3	24 h	3 days
UAE extension	every 30 min	4 h	3	6 h	12 h
Policy extension	every 60 min	24 h	2	24 h	48 h
Add a driver	every 60 min	24 h	2	24 h	48 h
Cancellation and refund	every 2 h	48 h	2	48 h	5 days

Per-insurer settings

- Channel: email, portal, or both
- Does the insurer need its own (underwriting) approval first?
- Is pricing one round or two?
- Email address per service, and the message template
- Does the round-two email need staff approval?

Price table with change history

- The price of each service at each insurer
- When a quoted price differs from the table, staff are alerted and an update is suggested
- No price changes without approval; the old price, who changed it and when are kept
- The alert also catches pricing mistakes on the insurer's side

SCREENS

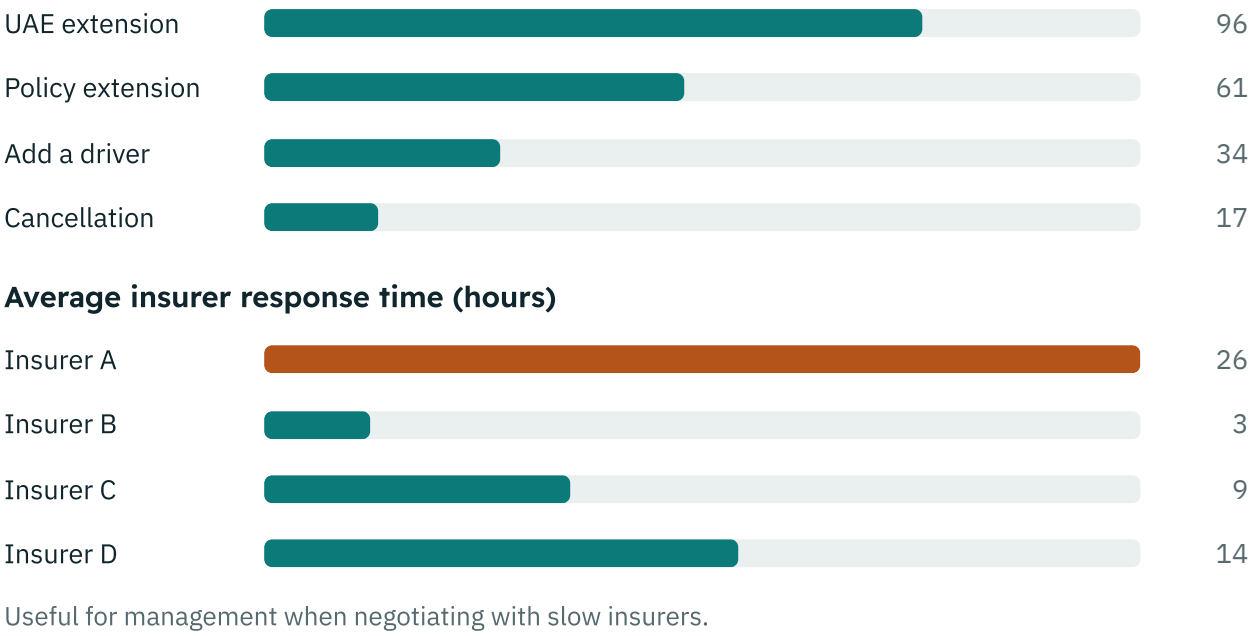
Stats dashboard

Tickets · this month — Illustrative figures

47 Open tickets	**9** In review (waiting on you)
18 Waiting on insurer	**312** Closed
64 Emails sent today	**11** Of which auto-reminders
1.6 Avg days to close	**4** Past deadline

Tickets by service type

Ownership transfer — 128



Useful for management when negotiating with slow insurers.

Staff approval screen

T-1042 · Ownership transfer · Insurer A

⏸ Approval 1 of 2: before sending to insurer

Data and documents

Current owner: **Khalid Al Balushi** New owner: **Salim Al Harthy** Plate: **12345 R B**

Policy: **MV-1880**

Vehicle registration · photo	✓ Matches
New owner's ID · PDF	✓ Valid
New owner's licence · photo	✓ Valid

Requirements source (RAG): "Insurer A procedures: ownership transfer, section 2".

What will be sent · via portal

Transaction: transfer of insurance to new owner
Policy number: MV-1880
New owner: Salim Al Harthy · civil ID from attached card
Attachments: 3 files
Requested transfer date: immediate

Approve and send **Edit** **Reject with note**

These screens are illustrative mock-ups. All names and numbers are fictional.

APPROACH

How it is built

Area	How Rasd covers it
AI-driven development	A `spec.md` is written before any code, with project rules for the AI coding assistant in `.cursor/rules`. AI then builds the code and tests under human direction, and every significant prompt and what was learned from it is recorded in `ai-dev-log.md`.
LangGraph	One graph per ticket with shared state, conditional branches (documents complete? approved or rejected?), loops (chasing missing documents), `interrupt` at both approval gates, and a checkpointer because a ticket can wait for days.
Engineering foundations	FastAPI and PostgreSQL, clear tables (customers, policies, tickets, messages, prices, approvals, audit log), screens for staff, settings and stats, pytest tests, Git, and Docker Compose to run everything with one command.
Cost reduction	The dashboard shows time saved: (manual time per ticket – staff time on the two approvals) × number of tickets, with the assumptions shown openly.

DELIVERY

Build phases

Phase 1

Foundation

spec.md, AI coding rules, Docker Compose (PostgreSQL and Temporal), schema, and sample data for customers, policies and 4 insurers.

Phase 2

Brain

The classification and extraction graph, document checks with OCR and validation rules, and a RAG knowledge base of insurer procedures.

Phase 3

Approvals

The two interrupt gates, the staff screen and the audit log.

Phase 4

Communication

Email (send, receive, match replies to tickets), a mock portal driven by Playwright, and a test-mode payment link.

Phase 5

Time

Temporal workflows for polling, reminders and escalation, all driven by per-service settings.

Phase 6

Showcase

The stats dashboard, settings and price screens, tests, and an end-to-end run of all three scenarios.

NOTES

Scope and things to watch

In the prototype

- Insurers, their portals and the payment link are all **simulated**, and customer data is fictional
- Email can run against a test inbox (such as Mailpit) so no real email leaves the system
- Connecting to real insurers is the next stage

In production

- Automating an insurer's portal with Playwright needs that insurer's agreement; an API from them is better
- Portal passwords live in a secrets vault, never in code
- Customer personal data is stored encrypted and is not sent to an external AI model without a data-protection agreement
- Any document read with low confidence goes to staff and is never accepted automatically